B. N. M. Institute of Technology

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Vidyayāmruthamashnuthe

Automated CI/CD Pipeline Implementation for Agile Task Management

Mini Kanban Board with Dockerized Deployment and Jenkins Automation

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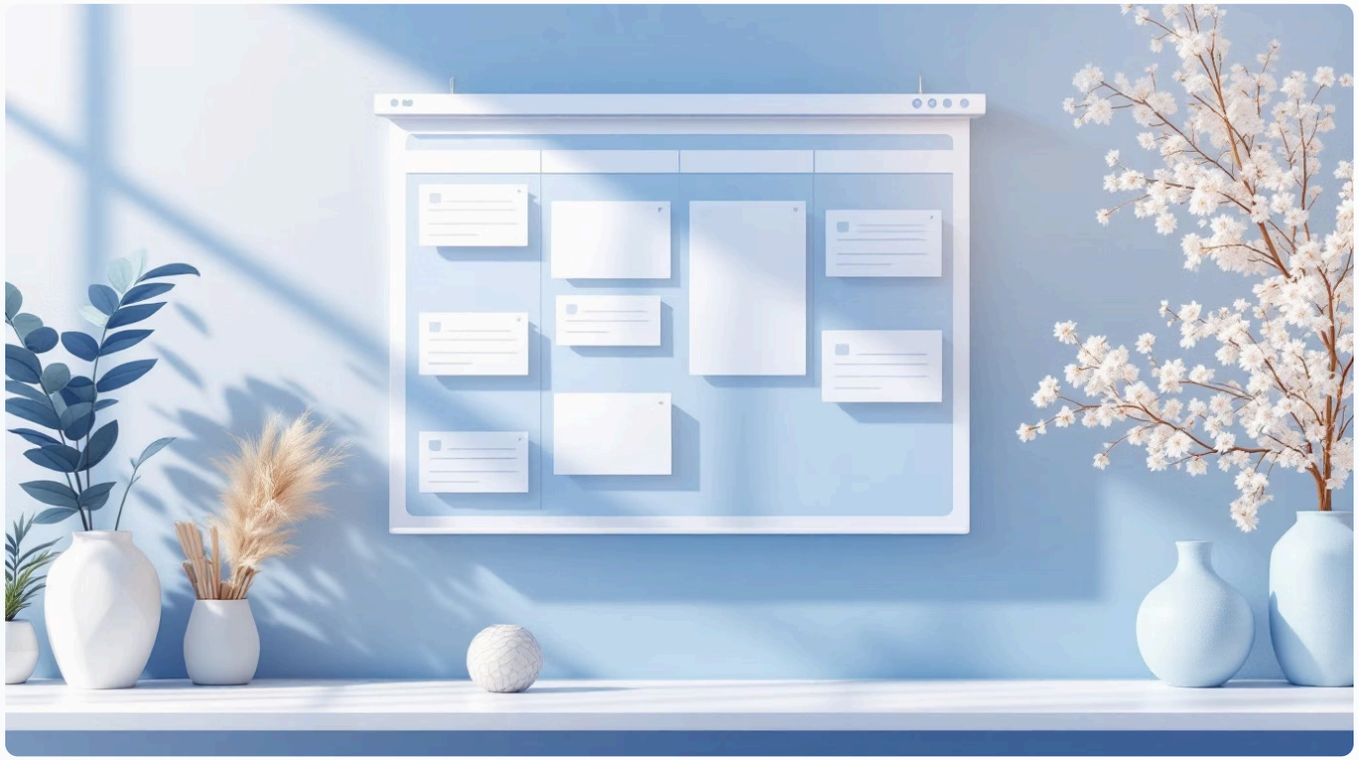
Project Overview

What We Built

A web-based Kanban board for Agile task tracking with three core columns: To-Do, In Progress, and Done. The system features an integrated CI/CD pipeline that automates testing and deployment while leveraging Docker-based containerization for consistent environments.



Key Capabilities

- Interactive drag-and-drop task management
- Real-time board updates
- Automated build, test, and deployment
- Separate development and production environments

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Problem Statement and Objectives

The Problem

Manual deployment processes are slow, error-prone, and lack consistency across environments. Without automation, teams face delays and deployment failures.

Our Objective

Automate the entire build, test, and deployment pipeline while ensuring code quality through automated testing and containerization for portability.

DevOps Focus

Implement CI/CD automation with fail-fast pipeline design and versioned releases to ensure reliability and rapid iteration.

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Technology Stack

Our implementation leverages modern tools and frameworks to deliver a robust, scalable solution.

Python Flask

REST API backend for routing requests, managing task workflows, and handling server-side application logic.

HTML/CSS/JS

Responsive frontend layer for rendering the user interface, capturing input, and communicating with the API.

Jenkins

CI/CD automation engine that runs build, test, and deployment pipelines with scripted release stages.

Docker

Containerization runtime used to package the application and its dependencies for consistent deployment across environments.

Git & GitHub

Source control and remote repository workflow for versioning code, coordinating collaboration, and triggering pipeline changes.

Pytest

Automated test framework for validating application logic, API behavior, and regression coverage within the pipeline.

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System Architecture

01

User Request

Frontend captures user actions and API calls

02

Frontend & API

UI layer processes input and sends to backend

03

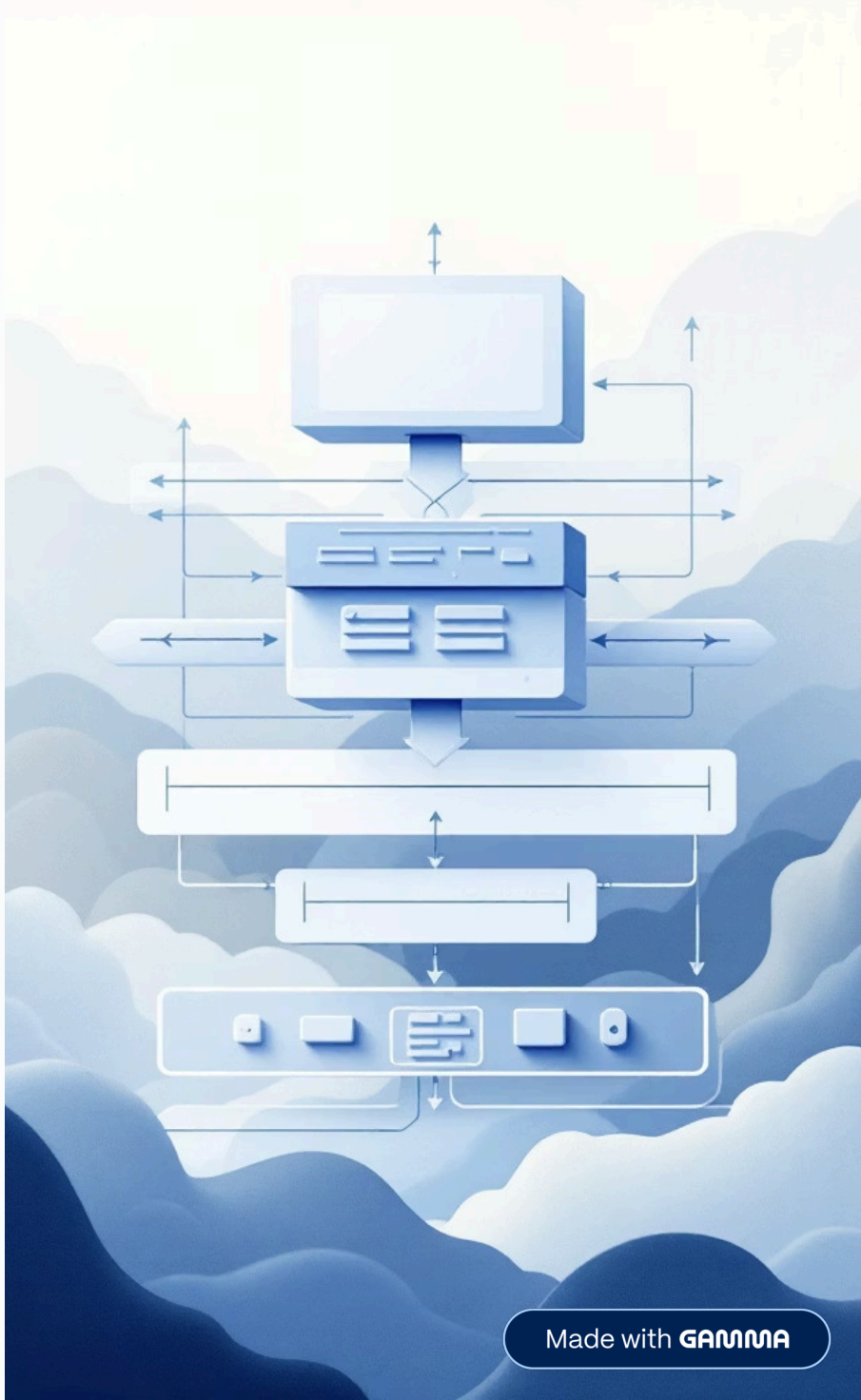
Backend Service

Business logic processes tasks and updates state

04

Data Persistence

State changes stored and UI refreshed



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CI/CD Pipeline Workflow

Our automated pipeline ensures quality at every stage, stopping deployment immediately if tests fail.

Clone Repository

Pipeline fetches latest code from GitHub

Install Dependencies

Set up Python environment and packages

Run Tests

Pytest validates functionality (Quality Gate)

Build Docker Image

Create container image with application

Run Container

Start application in isolated environment

Deploy Application

Launch to production or dev environment

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Key Features

Task Management

Create, update, and move tasks across To-Do, In Progress, and Done columns

Interactive Board

Drag-and-drop interface with real-time visual feedback

Status API

Monitor endpoint provides version and environment information

Automated Pipeline

Full CI/CD automation from code commit to deployment

Environment Isolation

Separate dev (Port 5001) and production (Port 5000) deployments

Real-Time Updates

Immediate UI reflection of task state changes

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Demo Flow and Output

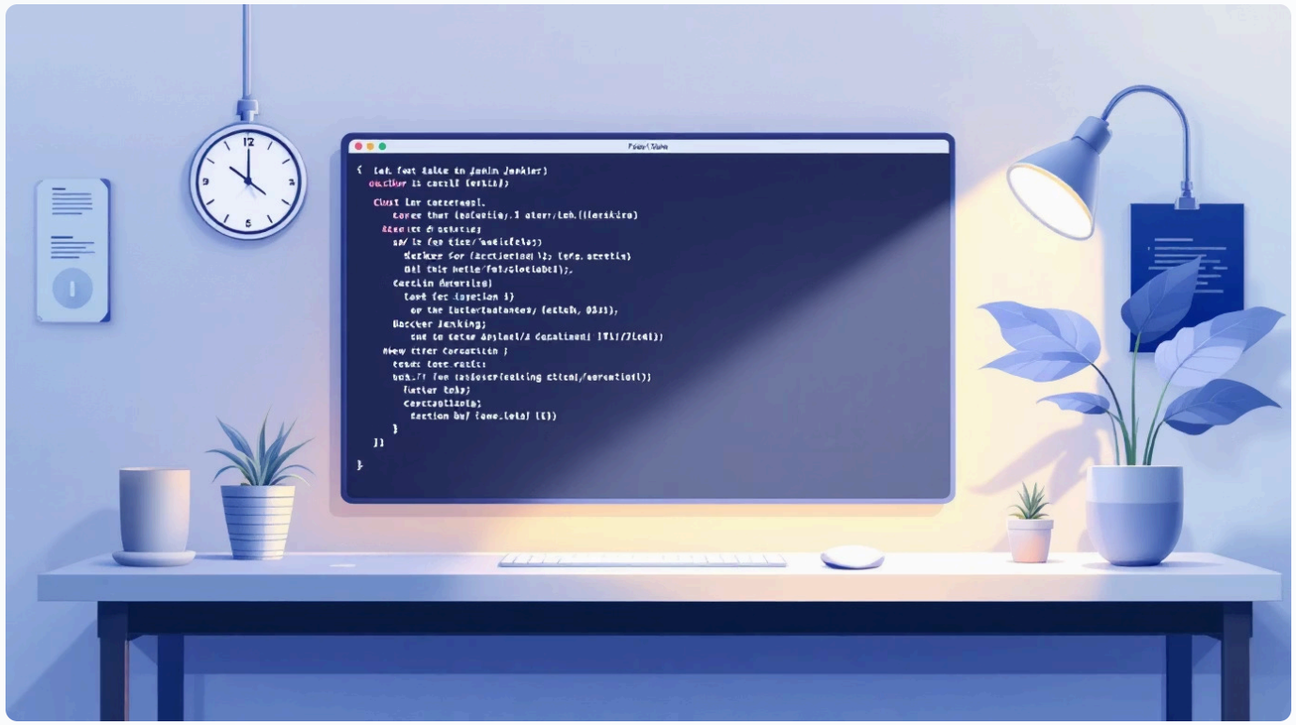
Execution Flow

1. User adds new task to board
2. Frontend sends API request
3. Backend processes and validates
4. Jenkins triggers pipeline
5. Docker builds and deploys
6. UI updates with new state

Example Outputs

Development: Port 5001
Production: Port 5000

Status endpoint returns JSON with version number, environment type, and uptime metrics.



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